Dispatch: MVP 2.0 Alpha (Assignment 4)

Group Rapid Route: Mihir Shinde, Ananya Das Hullumane Neelameghashyam · CS 549 · Live build: mihirshinde-29.github.io/rapid_route · Code: github.com/MihirShinde-29/rapid_route

Weekly schedule to the final

Week	Engineering	Assets	Milestone	Owner
W9 (Oct 26 – Nov 1), done	Emergency units in traffic; yielding (pull over, pass, merge back) and junction pre-emption; boids debug view; every building an address with its own entrance; door-to-door trips with left turns in and out; leisure trips; accidents and lane closures on one lane; dispatch core, scoring, difficulty, best scores; menu, HUD, end screen with response chart; random accidents; call alerts and edge arrows; reassigning returning units; shortcuts; ride along; coverage map; station placement; siren-aware unit ETAs	Ambulance, fire engine and police models; siren lights; station buildings; drawn entrances; block fill; siren audio, chime and UI sounds (all procedural)	MVP 2.0 integrated Oct 29, submitted Nov 2	Both
W10 (Nov 2 – 8)	Mid-crit feedback; mutual aid (send from further away when local units are busy); smarter turn-round for re-sent units; cars queued at a closure reroute	Incident animations (smoke, hazard lights, flares)	Play-test with 3+ outside players	Ananya: mutual aid, play-test; Mihir: turn-round, rerouting
W11 (Nov 9 – 15)	Multi-unit calls (crash: police and ambulance; fire: engine and ambulance); severity priority; ambulances take patients to hospital	Icons; sound; on-scene animations; model pipeline	Merge end of week	Ananya: multi-unit, severity; Mihir: hospital trips, sound, icons
W12 (Nov 16 – 22)	Cars reroute mid-trip on live costs; yielding tuned at junctions and on arterials; first-shift tutorial	Kenney cars and buildings; rain, fog; buses	Feature-complete build	Mihir: rerouting, yielding, vehicles; Ananya: tutorial, buildings
W13 (Nov 23 – 29)	Thanksgiving, light week: performance (1000 cars plus units), difficulty tuning, report experiments (yielding on/off, ETA accuracy, dispatch policy)	UI pass; pedestrians, parked cars; quality setting	Feature freeze Nov 29	Mihir: performance, experiments; Ananya: tuning, map
W14 (Nov 30 – Dec 6)	Could-have only if ahead: adaptive signals, two-lane arterials	Video, report figures, final slides	Release candidate Dec 6	Both
W15 (Dec 7 – 13): buffer	**No new features. Bug fixes, documentation, README, final report and video**		**Final submission**	**Both**

Release notes since A3 (Oct 13)

The game

- **Start:** menu over the live city with a how-to-play card. Mode (Dispatcher or Sandbox), shift (AM rush, midday, PM rush, night; 2 sim hours, played at 20×, about 6 real minutes) and difficulty. The best score for that choice is shown. Optional: move one station before the shift, with the coverage map previewing the result.
- **Active:** calls (medical, fire, crime, crash) come in at real addresses with a pop-up and a chime; off-screen calls get an arrow at the screen edge. Select a call, pick a unit from the list (ETA, fastest marked) or press Enter, and the unit drives there through traffic with its siren on. Units driving back can be sent again. Random accidents block a lane and bring crash calls, more at rush hour. Calls can also be placed by hand.
- **Sandbox:** free play like the A3 build, with no score or call list. Calls and accidents still bring units, sent automatically; place your own accidents, closures and calls and watch the response.
- **End:** the shift ends on time (win) or after too many missed calls (loss). The end screen shows a grade, score, calls handled and missed, mean response, a chart of each call's response time against its target, and the best score. Play again or return to the menu.
- **Difficulty:** Easy / Normal / Hard set calls per hour (12 / 20 / 30), missed calls allowed (4 / 3 / 2), units (Easy +1 ambulance per hospital, Hard −1 police car per station) and accident rate.

Simulation and pillars

- **Swarm and boids:** cars ahead of a unit pull onto the shoulder, the unit passes, they merge back; junctions ahead are held for it. A boids debug view (B) draws each unit's yield reach, held junctions and steering arrows. With yielding a unit crossed a test route in 120 s against 265 s stuck behind the same cars.
- **Pathfinding, made visible:** a coverage map (C) shades the city by how fast the nearest free unit could get there on live traffic (multi-source shortest paths). Unit ETAs now account for the siren: units arrive in a median 1.00 of their ETA at the morning peak.
- **City:** every building, park and car park is an address with its own entrance; trips run door to door, turning left on gaps in oncoming traffic. Accidents and lane closures block one lane; cars pass one side at a time. Unit and station models, drawn entrances, filler between buildings.
- **Also:** ride along with any unit; siren and UI sounds with mute; shortcuts with a help overlay; an error panel instead of a frozen page. From A3 feedback: the OD matrix as 9 districts with live counts, and incidents on any street with rerouting.
- **Stability:** 63 automated tests (Vitest): yielding, pre-emption, coverage against A*, unit ETAs against real trips, the call lifecycle, scoring, random accidents, and the morning peak (about 700 cars) with no gridlock.

Known issues

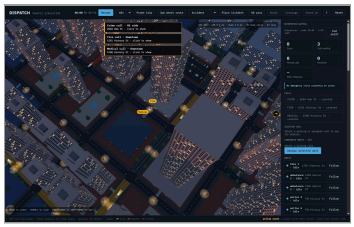
- **One lane per direction.** Making way for a siren is drawn as a 2 m move onto the shoulder; units pass only cars that have pulled over.
- **Lane closures are slow.** One side goes at a time, so a 65 m closure carries about 650–700 cars an hour; on a busy arterial at the peak, queues build. Cars reroute when the incident is placed, not while they wait (fix planned for W10).
- **A unit sent while driving back turns round at its next junction**, so its first few hundred metres can go the wrong way. Coverage counts it from that junction. Fix planned for W10.
- **Starting a shift reloads the page** at the shift's hour with half an hour of traffic run first. A station move is kept; incidents placed before are not.
- **Station placement** is one move per shift, onto a building or car park on a block laid out lot by lot.
- **At 300× the browser drops sim time** when it falls behind (a step takes about 0.5 ms at the peak).
- **Best scores are kept in this browser only.** Call pop-ups last 6 real seconds whatever the speed; at 300× only the newest 3 show (every call still has a pin and a list entry).
- **The gridlock guard** (removes a car stopped for 3 minutes) stays as a last resort; it removed none in the tests or in a sweep of 5 scenarios with 24 units each.
- **After an error the page has to be reloaded.**

Plan to the final

Full plan with work items, risks and cut order: `docs/FINAL_PLAN.md`. Feature freeze Nov 29, release candidate Dec 6.

- **Deeper game (W10–W11):** mutual aid, multi-unit calls, severity (Ananya); ambulances take patients to hospital (Mihir).
- **Smarter traffic (W10–W12, Mihir):** sensible turn-round, cars queued at a closure reroute, mid-trip rerouting on live costs with a margin so routes don't flip, yielding tuned at junctions.
- **Livelier city (W10–W13):** animated incidents (smoke, hazard lights, flares) and on-scene work (flames, hose spray, police tape). CC0 low-poly models from Kenney: vehicles with spinning and steering wheels, body pitch and roll, doors opening on scene (Mihir); buildings fitted to lots by site kind, with lit windows and fire damage (Ananya). Rain and fog that slow traffic, buses, pedestrians, parked cars. A quality setting keeps the current models as a fallback.
- **Evidence (W13):** experiments over many city trips: yielding on vs off, ETA accuracy per shift, dispatch by distance vs ETA, coverage before and after a station move.
- **Finish (W12–W15):** tutorial, icons, sound mix, night lighting, 1000 cars at 60×, video and final report.
- **If late,** the visuals are cut first (pedestrians, buses, Kenney buildings); mutual aid, multi-unit calls, rerouting and the experiments stay.

Screenshots



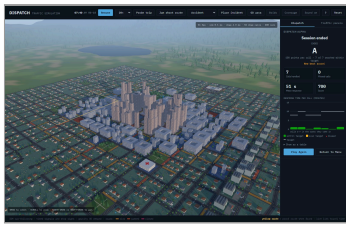
Call alerts: pop-ups, pins over calls in view, an arrow to an off-screen call.



Ride along with an ambulance on its way to a medical call.



Coverage map: time for the nearest free unit, on live traffic.



End screen: grade, score, response time per call against its target.